Cross-Scale Style-Guided Enhancement for Underwater Remote Sensing Imagery

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***Abstract*—Underwater imaging is essential for marine remote sensing tasks, such as environmental monitoring, resource exploration, and autonomous navigation. However, images captured in underwater environments often suffer from complex degradations, including wavelength-dependent color distortion, contrast attenuation, and structural detail loss. To address these challenges, we propose a cross-scale style-guided network (CSG-Net) for robust underwater image enhancement (UIE). CSG-Net employs a dual-stage collaborative framework that decouples global degradation modeling from local detail refinement. In the first stage, a style extraction network (SE-Net) extracts multiscale degradation-aware style priors that implicitly encode large-scale physical degradation patterns, such as red-channel attenuation and spectral imbalance. In the second stage, a style-guided enhancement network (SG-Net) leverages these style features to guide spatially adaptive enhancement, enabling consistent color correction and fine-grained detail recovery. To alleviate semantic degradation during scale transitions, CSG-Net introduces the proposed multiresolution feature-preserving cross-scale interaction (MFPCSI) module, which enhances the preservation and integration of hierarchical features. Combined with the multistream information fusion (MSIF) module, this design enables the effective fusion of semantic and structural information across spatial scales. The proposed components enable the preservation of fine-grained details while adaptively integrating semantic and structural cues across multiple scales. Comprehensive experiments conducted on diverse and challenging underwater image datasets demonstrate that CSG-Net consistently surpasses state-of-the-art approaches in terms of the peak signal-to-noise ratio (PSNR), the structural similarity index measure (SSIM), and the underwater image quality measure (UIQM). Furthermore, the model exhibits strong cross-domain generalization and delivers high-fidelity visual results, underscoring its suitability for deployment in practical vision systems operating under complex, real-world environments.**

***Index Terms*—Cross-scale feature fusion, style-guided restoration, underwater imaging, underwater remote sensing.**

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I. Introduction

UNDERWATER imaging has emerged as a critical component of marine remote sensing, enabling detailed observation, monitoring, and mapping of submerged environments such as coral reefs, seafloor habitats, and underwater infrastructures. However, due to the combined effects of wavelength-dependent light absorption, scattering phenomena, and multipath transmission in aquatic environments, captured underwater images often suffer from severe color distortions, contrast degradation, and loss of fine details [1], [2], [3]. These degradation effects significantly impair the performance of downstream visual tasks, such as underwater object detection, scene reconstruction, and autonomous navigation, posing substantial limitations on the reliable application and further development of underwater remote sensing technologies [4], [5], [6], [7].

To better understand the physical mechanisms underlying underwater image degradation, numerous studies have established physical imaging models, among which the Jaffe–McGlamery model is one of the most representative. This model, based on the principles of light propagation, quantitatively characterizes the attenuation and scattering effects that lead to energy loss and image quality degradation, thus providing a theoretical foundation for early underwater image enhancement (UIE) and restoration methods [8], [9], [10], [11]. By estimating parameters such as background light and transmission rate, these models can partially recover image quality under specific conditions, demonstrating the importance of physical modeling in underwater imaging.

In Fig. 1, the Jaffe–McGlamery model [12] decomposes the underwater imaging process into three components: direct attenuation, forward scattering, and backward scattering. Among these, forward scattering is generally negligible under

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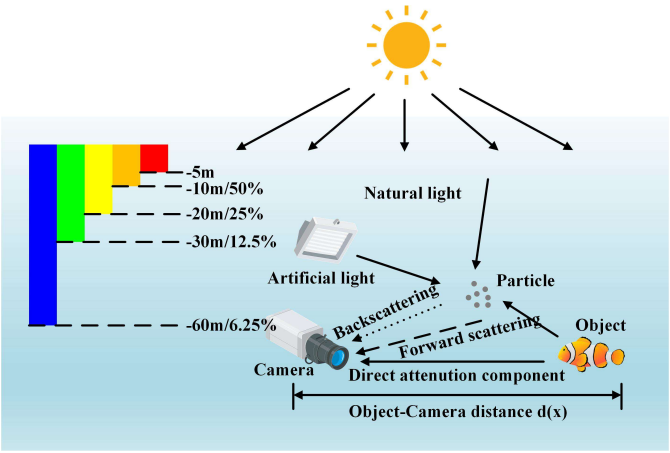

Fig. 1. Illustration of the underwater imaging process, including direct attenuation, forward scattering, and backward scattering, along with the selective absorption characteristics of light.

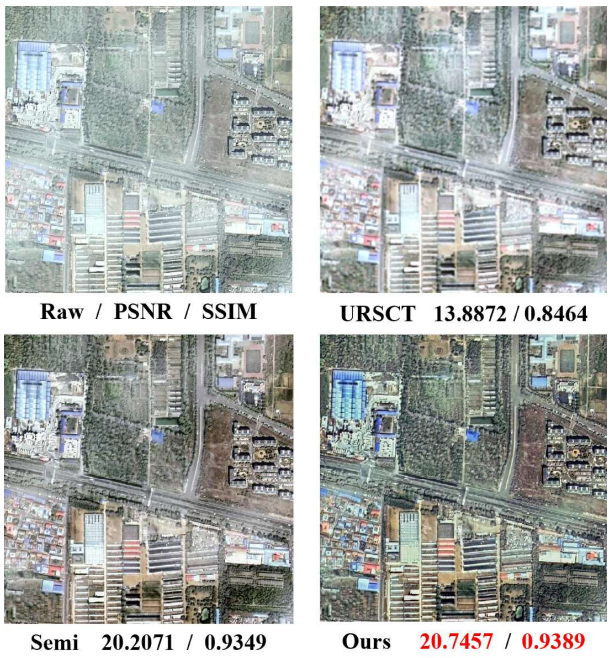

Fig. 2. Visual comparison on remote sensing images with SOTA methods. Our method achieves superior detail restoration and the highest PSNR and SSIM metrics.

most practical conditions. Therefore, the simplified underwater imaging model can be formulated as follows:

$$I_\lambda = J_\lambda T_\lambda + A\,(1 - T_\lambda) \tag{1}$$

where $\lambda \in \{R, G, B\}$, I denotes the degraded underwater image captured by the camera, J represents the clear image without degradation, T is the transmission map, and A denotes the global background light. In this formulation, $J_\lambda T_\lambda$ corresponds to the direct attenuation component, which primarily causes contrast reduction and color distortion, while $A(1 - T_\lambda)$ represents the backward scattering component, which leads to blurriness and haziness in the image. Specifically, T_λ represents the degree of light attenuation in the water, which is closely related to the water's turbidity, the path length of light, and the scattering and absorption properties of the water

$$T_\lambda = \exp\left(-\beta_\lambda d\right) \tag{2}$$

where $\exp(\cdot)$ is the natural exponential function, β_λ denotes the attenuation coefficient in λ, and d represents the distance from the camera to the target object.

Although the aforementioned imaging models enable partial recovery of underwater image quality, their performance is substantially limited in complex, real underwater environments. The high heterogeneity and dynamic variability of water conditions, caused by variations in water types, suspended particle concentrations, and artificial lighting interferences, make it extremely challenging to accurately estimate the model parameters or comprehensively describe the degradation process. Consequently, traditional model-based approaches often suffer from insufficient adaptability and limited enhancement performance in practical applications.

In recent years, data-driven deep learning methods have emerged as a promising alternative for visual enhancement results in complex environments, as partially illustrated in Fig. 2. By directly learning the mapping between degraded and enhanced images, these approaches alleviate the dependence on precise physical parameters and partially overcome the limitations of traditional methods. However, existing learning-based models generally struggle to simultaneously maintain global structural consistency and restore local fine details, particularly under conditions of complex and dynamic degradation. This often results in oversmoothed outputs or amplified noise artifacts, thereby hindering further improvements in enhancement performance [13].

A comprehensive analysis reveals that achieving effective low-quality image enhancement under complex environmental conditions faces several critical challenges: 1) the difficulty of accurately modeling multifactor degradation processes induced by diverse optical properties of water bodies; 2) the inherent conflict between maintaining global content consistency and recovering local fine-grained details; and 3) the semantic and structural information loss during multiscale feature fusion, which restricts the reconstruction of fine-grained details.

To address the aforementioned challenges, we propose a novel CSG-Net for robust underwater imaging in complex environments. CSG-Net adopts a dual-stage architecture: the first stage extracts holistic style features at a lower resolution to guide global color correction and structural consistency restoration, achieving a balance between perceptual quality and computational efficiency; the second stage refines local textures and fine-grained structures under the guidance of the extracted style priors, effectively balancing global consistency and local detail richness. Furthermore, to mitigate semantic and structural information loss during multiscale feature interactions, CSG-Net integrates a multiresolution feature-preserving cross-scale interaction module (MR-N) and a multistream information fusion (MSIF) module, which dynamically consolidates multiscale features to enhance fine-grained structural restoration under complex underwater degradation.

The main contributions of this article are summarized as follows.

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1) We propose a dual-stage UIE framework that decouples global structural restoration and local detail recovery, enabling task-specific optimization under complex degradation scenarios.
2) We introduce a physically inspired style-guided enhancement strategy, where style priors learned by SE-Net implicitly encode wavelength-dependent degradation patterns (e.g., color shift and contrast loss) and guide structure-consistent restoration in SG-Net.
3) We design a multiresolution-preserving cross-scale interaction mechanism based on MFPCSI and an MSIF module, which jointly enhances semantic consistency and texture fidelity during multiscale feature fusion.
4) We conduct comprehensive experiments across multiple public underwater image datasets (UIEB, UCCS, and U45), demonstrating that CSG-Net outperforms state-of-the-art methods in both full- and no-reference metrics and exhibits robust generalization across diverse environmental conditions.

The remainder of this article is organized as follows. Section II reviews UIE techniques related to this work, covering both traditional physical modeling approaches and recent data-driven methods and systematically analyzes the limitations and development trends. Section III provides a detailed description of the proposed cross-scale style-guided network (CSG-Net), including the overall framework design, the dual-stage collaborative enhancement mechanism, and the specific implementation of the multiresolution feature-preserving cross-scale interaction (MFPCSI) and the MSIF module. Section IV presents the experimental setup and evaluation protocols, including dataset descriptions, evaluation metric definitions, and comparative method settings, followed by comprehensive quantitative and qualitative results on multiple public underwater image datasets to verify the effectiveness and robustness of the proposed approach. Finally, Section V concludes this article by summarizing the main contributions, discussing current limitations, and outlining possible directions for future research.

II. Related Work

In this section, we review approaches closely related to the proposed method, with a particular focus on UIE techniques.

A. Underwater Image Enhancement

In recent years, numerous methods have been developed to improve the quality of underwater images. These approaches for UIE can be broadly categorized into traditional methods and deep learning-based methods. Traditional methods are generally divided into physics- and nonphysics-based approaches, depending on whether they explicitly model the physical degradation processes in underwater environments.

Physics-based methods aim to model the optical degradation mechanisms that occur during underwater imaging. These methods typically estimate background light and transmission maps using various priors to restore degraded images. For instance, Drews et al. [14] proposed the underwater dark channel prior (UDCP), which restores underwater image degradation based on the selective absorption of light by water. Peng and Cosman [15] introduced the image blurriness and light absorption (IBLA) method, which restores underwater images by exploiting the properties of IBLA before estimating scene depth. Song et al. [16] presented a fast depth estimation method using the underwater light absorption prior (ULAP). Berman et al. [17] proposed the haze-line model to address wavelength-dependent attenuation. Zhou et al. [18] developed a comprehensive imaging model that dynamically estimates scene depth from a single image, eliminates backscatter across different depth ranges, and applies color correction for low-light enhancement. Despite their effectiveness, these physics-based methods heavily rely on manually crafted priors, which limits their adaptability to diverse underwater conditions.

Nonphysics-based methods enhance underwater images directly through pixelwise operations without explicitly modeling the physical imaging process. Ancuti et al. [19] introduced a fusion-based approach that aggregates multiple feature maps with weighted blending. Subsequently, Ancuti et al. [20] proposed a multiscale fusion technique that significantly improves color balance, particularly in the red channel of underwater images. Zhuang et al. [21] proposed a Bayesian Retinex framework that decomposes enhancement into two independent denoising problems using priors on reflectance and illumination at multiple scales. Recently, Zhou et al. [22] presented a method based on multi-interval subhistogram perspective equalization to improve contrast and color fidelity. However, nonphysics-based enhancement methods often disregard the underlying optical properties of underwater scenes, which may lead to overenhancement and overexposure artifacts.

Deep learning-based methods have recently emerged as a promising direction for UIE, leveraging the strong representation capabilities of convolutional neural networks. Li et al. [23] proposed UWCNN, a UIE network trained on synthetically generated underwater datasets. However, its simple network structure limits the robustness across different types of water. Li et al. [24] introduced WaterGAN, an unsupervised color correction network that integrates a generative adversarial network (GAN) with a physical underwater imaging model to synthesize enhanced images and depth maps. Li et al. [25] further proposed WaterNet, a gated fusion network, and established the UIE benchmark dataset (UIEBD). Zhou et al. [26] developed a multiview UIE method through an embedded fusion mechanism. Zhang et al. [27] introduced an enhancement network guided by extreme scene reflectance priors. Zhou et al. [28] proposed UGIF-Net, a lightweight, fully guided information flow network that incorporates multiple guidance cues for underwater enhancement. Although deep learning-based methods have achieved notable progress, they still face challenges due to the scarcity of large-scale, high-quality paired training datasets and often exhibit limitations in generalization to diverse underwater environments.

Transformer-based methods have emerged as powerful tools for image enhancement due to their ability to model long-range dependencies. In the context of UIE, several Transformer-based frameworks have been proposed. Peng et al. [29] proposed a U-shape Transformer, which

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employed a U-shaped architecture that integrated channel-wise multiscale feature fusion and spatialwise global feature modeling, effectively addressing wavelength-dependent attenuation across color channels and spatial regions. Shen et al. [30] proposed UDAformer by introducing a dual-attention Transformer framework, in which the dual-attention Transformer block combines channel and pixel self-attention, and a shifted-window mechanism improves computational efficiency. Li et al. [31] leveraged a physically inspired Swin Transformer, jointly exploiting CNNs and Transformers to capture local and global dependencies while embedding physical degradation priors to mitigate the lack of inductive bias in limited datasets, achieving competitive results across multiple benchmarks. Shang et al. [32] presented a luminance-guided transformer (LGT) framework, with key innovations including the underwater scene enhancement curve for adaptive luminance balancing and wavelength compensation, as well as the luminance-guided self-attention for dynamic multifeature fusion, effectively enhancing details while preserving global consistency. Qing et al. [33] proposed HG2former, which incorporated HSV color modeling and nonlinear gamma correction to enhance hue, saturation, value, and contrast separately; HG2former encoder–decoder design with improved multihead self-attention captured both local and long-range dependencies, enabling more precise color correction and luminance adaptation. Han et al. [34] proposed a unified CNN-Transformer (UMCTN) framework tailored for UIE. However, Transformer-based methods rely on self-attention, whose computational complexity scales quadratically with image feature sizes, limiting their deployment in resource-constrained environments.

B. Style-Guided Feature Modeling

Style-guided learning has attracted increasing attention for its ability to capture domain-specific appearance priors beyond pixel-level constraints, enabling models to preserve structural semantics while adapting to diverse visual domains. Gatys et al. [35] introduced style-guided learning by disentangling content and style representations in CNNs, showing that deep layers capture coherent visual styles. Gatys et al. [35] established the foundation for subsequent style-based image generation and translation methods. Fan et al. [36] proposed the style-guided U-Net (SGU-Net) for medical image super-resolution. SGU-Net employs a series of style-guided U-Blocks (SUBs), where each block comprises a main branch for multiscale feature aggregation and a mask branch that adaptively reweights feature maps based on estimated style relevance, thereby enhancing structural fidelity and style-aware reconstruction.

To address the mode collapse problem, Choi et al. [37] introduce a style-guided discriminator with a flexible decision boundary and disentangled representation learning. The framework encourages domain-independent feature encoding and adaptive style alignment, effectively improving mode diversity and translation quality across multiple datasets, including CelebA-HQ, AFHQ, and Yosemite. Mao and Ma [38] constructed a visual style metric using Gram-matrix-based feature distances and employed a triplet loss with adaptive margins to guide style embedding. By jointly optimizing generated and reference image pairs in a learned metric space, the approach effectively improves cross-domain style alignment without altering the original network architecture. Zhao et al. [39] proposed OmniStyleGAN, which employs omnidimensional dynamic convolutions to enhance style modeling and feature adaptability in image-to-image translation. OmniStyleGAN incorporates a specialized style encoder for style-guided synthesis, enabling a more effective capture of structural and textural characteristics in the input images.

While existing style-guided frameworks primarily focus on visual or domain-level consistency, they often overlook structural fidelity and representational coherence across scales. To address this, we introduce CSG-Net, which reinforces cross-scale consistency between degradation-aware style priors and structural representations, achieving perceptually coherent enhancement in complex underwater scenes.

III. Physically Consistent Style-Guided Enhancement Method

To address the challenges of modeling complex underwater degradations, balancing global consistency with local detail enhancement, and mitigating semantic loss during multiscale feature fusion, we propose a novel dual-stage framework termed CSG-Net. The proposed architecture explicitly decouples structural restoration and detail enhancement by incorporating a style-guided mechanism that facilitates efficient feature reuse and semantically consistent guidance.

CSG-Net consists of two successive stages: the style extraction network (SE-Net) and the style-guided enhancement network (SG-Net). The first stage extracts global style representations under multiscale semantic fusion to model the degradation characteristics of underwater scenes. These style priors are then injected into the second stage to guide the enhancement process, enabling local texture refinement and global perceptual consistency. To further improve the quality of multiscale feature representation, CSG-Net integrates the MSIF module and a multiresolution fusion backbone inspired by MFPCSI, ensuring robust structural alignment across scales. The overall framework is illustrated in Fig. 3. In Sections III-A–III-D, we present a detailed description of each core component.

A. Overall Network Architecture

As illustrated in Fig. 3, the proposed CSG-Net adopts a dual-stage architecture to explicitly address the conflict between global structural consistency and local detail restoration in UIE. The network consists of two key components: the SE-Net and the SG-Net.

The first stage, SE-Net, is designed to extract global degradation-aware style priors that explicitly capture the dominant degradation factors in underwater environments, including channel-dependent color bias, reduced contrast, and structural distortions. Specifically, SE-Net employs a pretrained EfficientNet-B0 backbone to obtain multiscale semantic features with minimal computational overhead. These features are refined and fused using the MFPCSI strategy, which preserves spatial fidelity across scales. The fused

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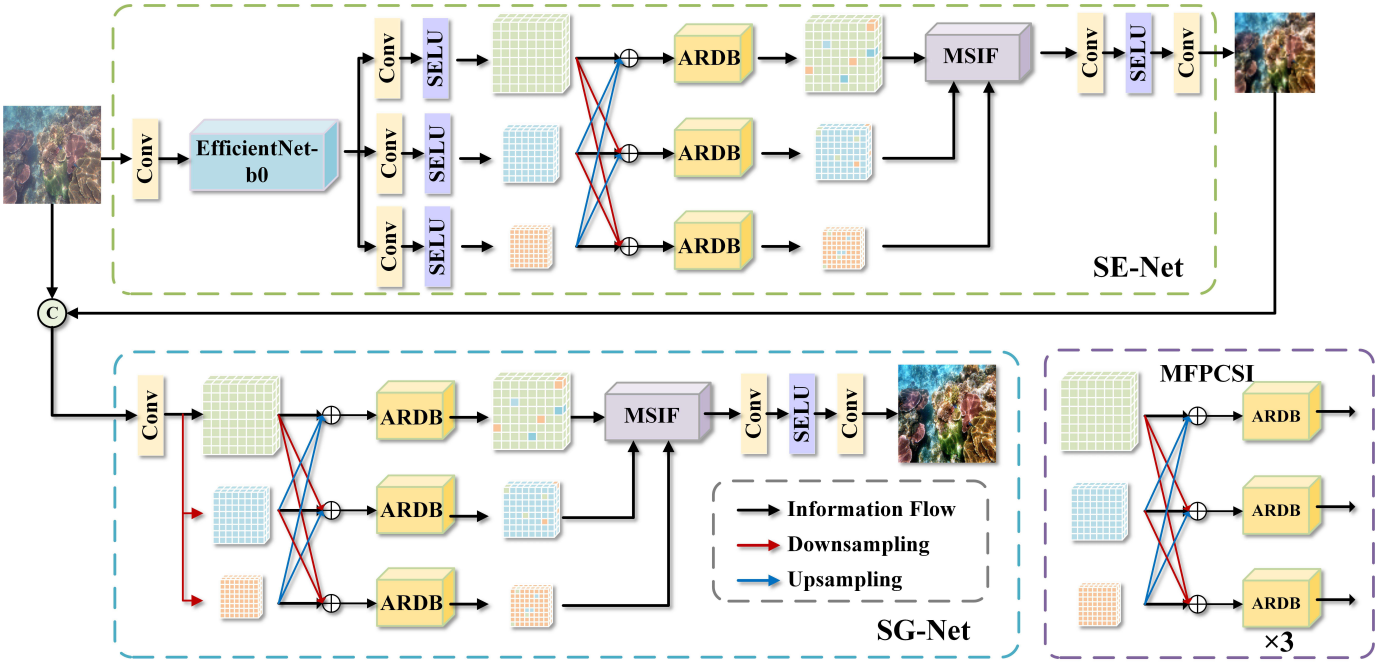

Fig. 3. Overview of the proposed CSG-Net framework. The architecture consists of two stages: the SE-Net and the SG-Net. SE-Net extracts multiscale features using EfficientNet-B0, which are further fused through MFPCSI and the proposed MSIF module to obtain the style prior. The style prior is used in SG-Net to guide the detail enhancement of the degraded input image.

multistream representations are then integrated through the MSIF module, producing a low-dimensional feature vector that encodes global color distribution statistics, contrast variations, and structural degradation cues in a quantitatively interpretable manner.

The second stage, SG-Net, focuses on detail enhancement and visual fidelity improvement under the guidance of the extracted style priors. Specifically, the original degraded image is concatenated with the style representation from SE-Net, forming a fused input that is processed through a similar architectural backbone comprising EfficientNet-B0, MFPCSI, and MSIF. This stage enables fine-grained detail recovery in color-suppressed and texture-degraded regions while maintaining structural alignment with the global style constraints.

This two-stage collaborative framework enables a clear functional decoupling of global and local enhancement tasks and facilitates parameter sharing and computational efficiency. By reusing the backbone design across both stages, CSG-Net minimizes redundancy while preserving critical semantic and spatial cues across the enhancement pipeline.

In the following sections, we provide detailed explanations of the SE-Net, SG-Net, and MSIF modules, as well as the training objective and loss design.

B. Style Extraction Network

The SE-Net is designed to capture global style representations that characterize the underlying degradation patterns of underwater images. These style features serve as high-level priors for guiding the subsequent enhancement stage, especially in restoring content structure and compensating for global color shifts.

EfficientNet [40] introduces a framework for systematically scaling convolutional neural networks. Instead of independently adjusting network width, depth, or input resolution in an ad hoc manner, EfficientNet employs a compound scaling strategy that uniformly scales all dimensions using a simple set of fixed coefficients. This structured scaling approach, combined with AutoML techniques for architecture search, enables EfficientNet to achieve significantly improved computational and parameter efficiency compared to conventional CNN architectures.

To achieve an optimal balance between computational efficiency and feature representation capacity, we adopt the pretrained EfficientNet-B0 as the backbone for multiscale feature extraction in the proposed CSG-Net. EfficientNet employs depthwise separable convolutions, which significantly reduce computational complexity while preserving strong representational power—making it particularly suitable for UIE tasks constrained by efficiency requirements.

Specifically, we utilize the first eight layers of EfficientNet-B0 and extract intermediate feature maps after the layers indexed at 2, 4, and 7, corresponding to spatial resolutions of 1/4, 1/8, and 1/16 of the input image size, respectively. This hierarchical representation allows the network to effectively capture both low-level spatial details and high-level semantic context, facilitating robust enhancement under complex underwater imaging conditions.

The extracted features at these three scales are subsequently fed into a multiresolution fusion based on MFPCSI. Unlike conventional sequential downsampling, MFPCSI maintains parallel resolution branches and performs repeated multiresolution fusion, thereby preserving spatial structure and semantic consistency across multiscales.

To consolidate these features into a unified style representation, we introduce the MSIF module. MSIF aggregates the outputs from different MFPCSI branches and performs

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scale-aligned fusion at the 1/4 resolution level. The result is a compact style descriptor that emphasizes content-level degradation patterns while suppressing local noise and unstable texture responses.

It is worth noting that SE-Net operates entirely at 1/4 resolution or lower, which significantly reduces the parameter count and computational complexity. More importantly, the extracted style representation focuses on the holistic appearance of the input rather than fine-grained textures, providing a strong global prior for the detail enhancement process in SG-Net.

In underwater environments, the dominant degradation phenomena, such as red channel attenuation, bluish-green color shift, and global contrast suppression, exhibit spatially coherent patterns due to the wavelength-dependent absorption and scattering properties of water. These global degradations can be effectively abstracted as the style of the underwater scene. SE-Net, through its multiscale fusion strategy and multiresolution feature preservation, implicitly captures these degradation-aware style features without relying on explicit physical modeling. The resulting style representation encodes the scene's spectral imbalance and structural degradation tendencies, enabling this representation to serve as a powerful prior that enhances the semantic consistency of the enhancement process and facilitates efficient guidance in SG-Net for region-specific texture recovery under complex underwater conditions.

C. Style-Guided Enhancement Network

The SG-Net serves as the second stage of the CSG-Net framework, focusing on refining local textures and perceptual details under the guidance of the global style representations extracted in the first stage. Specifically, while SE-Net encodes global degradation priors associated with wavelength-dependent absorption and contrast attenuation, SG-Net leverages these priors to adaptively enhance the scene with spatial awareness, thereby achieving more faithful texture restoration and perceptual quality improvement.

Specifically, SG-Net takes two inputs: the degraded underwater image and the degradation-aware style descriptor generated by SE-Net. To ensure spatial alignment, the style representation is first upsampled via nearest-neighbor interpolation to match the spatial dimensions of the original image. These two inputs are then concatenated along the channel dimension, forming a fused representation that jointly encodes both global degradation trends and raw pixel-level observations.

This concatenated input is passed through a two-level downsampling process to obtain multiresolution feature maps at 1/2 and 1/4 scales. These features are subsequently processed by the multiresolution preserving MFPCSI, which performs repeated cross-scale fusion while maintaining spatial localization. The MFPCSI is followed by the MSIF module to aggregate features from different resolutions and enhance semantic richness and detail integrity.

By conditioning the enhancement process on a physically meaningful style prior, SG-Net allows the model to focus computational resources on regions most affected by underwater degradation, such as red-depleted, low-contrast, or hazy areas, while maintaining global consistency in color and structure. This guided refinement improves local texture recovery and ensures that the enhanced image aligns with the physical plausibility and perceptual realism expected in underwater environments. SG-Net, paired with SE-Net, forms a synergistic dual-stage framework that balances global scene restoration with fine-grained detail enhancement, achieving a unified, style-consistent enhancement of underwater images across diverse degradation conditions.

D. Multiresolution Feature-Preserving Cross-Scale Interaction

To effectively capture and integrate multiscale contextual features, we employ MFPCSI as the backbone for feature fusion within both SE-Net and SG-Net. As shown in Fig. 3, MFPCSI maintains multiresolution representations throughout the entire network by connecting parallel convolutional branches at different resolutions. This architecture enables rich semantic encoding at coarse scales while preserving spatial accuracy at finer resolutions, which is particularly critical in underwater imaging, where detail degradation and global color shifts often coexist.

In underwater scenes, objects and textures may appear at vastly different scales, ranging from distant seabed structures to nearby fine textures, such as marine organisms or artificial targets. Conventional encoder–decoder networks suffer from resolution loss during downsampling, leading to spatial inconsistencies. MFPCSI mitigates this issue by concurrently maintaining multiple resolution streams and performing iterative fusion, which ensures structural consistency across scales and significantly improves depth perception.

To further enhance the semantic and structural representation capability of MFPCSI, we integrate an attention-refined dense block (ARDB) at each fusion stage. As illustrated in Fig. 3, ARDB modules are inserted after feature aggregation to refine the fused outputs by adaptively enhancing salient features and suppressing irrelevant or noisy signals. The ARDB structure is detailed in the next subsection.

The input to MFPCSI comprises three parallel streams with spatial resolutions of 1/4, 1/8, and 1/16 of the original input. At each fusion stage, every branch is followed by an ARDB, which selectively consolidates salient cues and propagates them across scales. MFPCSI then performs three successive cross-scale fusion-and-propagation stages. Through repeated exchanges among the branches, high- and low-resolution information is jointly modeled and mutually reinforced so that the output of each branch becomes a multistream representation that aggregates features from all scales. This design enables more effective information integration and leads to sharper details and improved overall visual quality in the enhancement results. Formally, the equations can be written as

$$i_1 = \mathrm{ARDB}\left(f_1 \oplus U\left(f_2\right) \oplus U\left(U\left(f_3\right)\right)\right) \tag{3}$$

$$i_2 = \mathrm{ARDB}\left(D\left(f_1\right) \oplus f_2 \oplus U\left(f_3\right)\right) \tag{4}$$

$$i_3 = \mathrm{ARDB}\left(D\left(D\left(f_1\right)\right) \oplus D\left(f_2\right) \oplus f_3\right) \tag{5}$$

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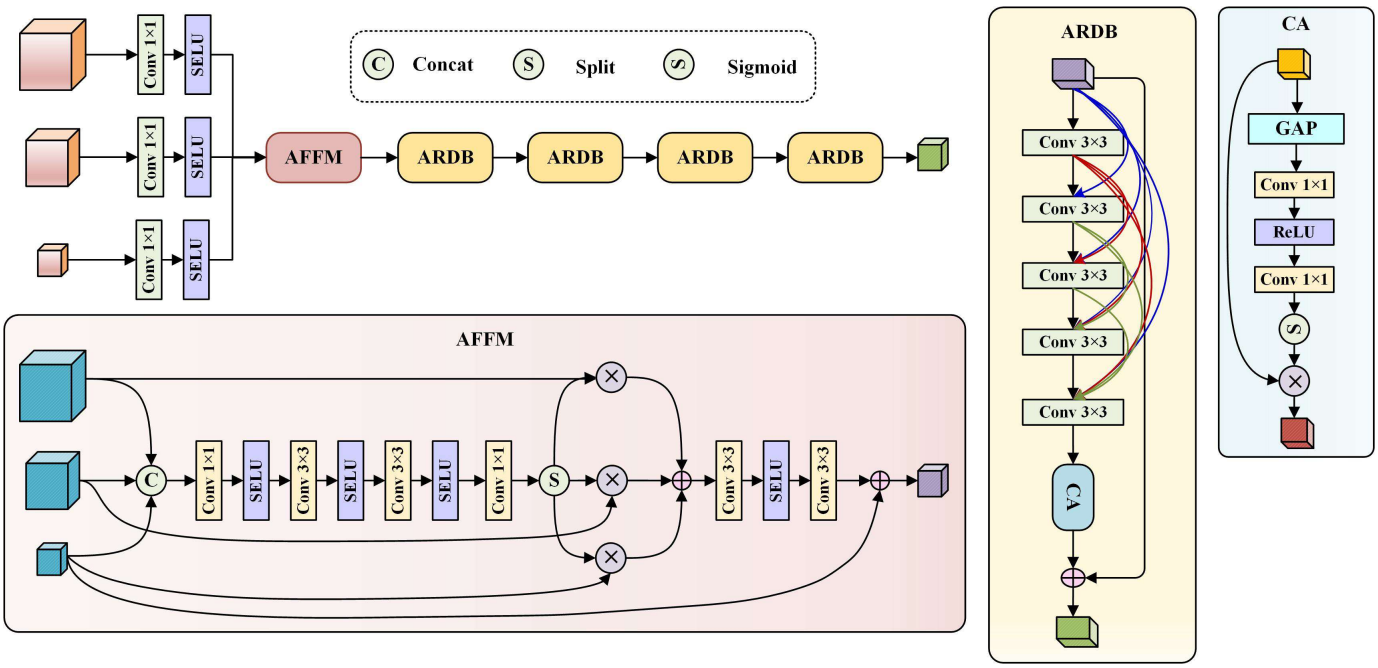

Fig. 4. Architecture of the proposed MSIF module. MSIF consists of two key submodules: the AFFM, which adaptively aggregates multiscale features by learning spatial-channel dependencies, and the ARDB, which enhances local details through residual-dense connections and channel attention. Together, they enable effective integration and refinement of hierarchical representations for detail-preserving enhancement.

where i_1, i_2, and i_3 denote the multistream outputs from the high-, middle-, and low-resolution branches (with spatial resolutions of 1/4, 1/8, and 1/16 of the input), respectively. The detailed design of ARDB is shown in Fig. 4. f_1, f_2, and f_3 represent the high-, middle-, and low-resolution features, respectively. The operator $\oplus$ indicates elementwise addition, while $U(.)$ and $D(.)$ denote upsampling and downsampling operations, respectively.

Upsampling $U(\cdot)$ is implemented via nearest-neighbor interpolation, which assigns each upsampled pixel the value of its nearest source pixel and, thus, avoids the edge blurring commonly introduced by bilinear interpolation. Downsampling $D(\cdot)$ is defined as

$$y = \mathrm{ReLU}\left(\mathrm{Pool}_{\mathrm{avg}}\left(\mathrm{Conv}\left(\mathrm{ReLU}\left(\mathrm{Conv}\left(x\right)\right)\right)\right)\right) \tag{6}$$

where x and y denote the input and output variables, respectively; $\mathrm{ReLU}(\cdot)$ is the rectified linear unit activation; $\mathrm{Pool}_{\mathrm{avg}}(\cdot)$ denotes average pooling; and $\mathrm{Conv}(\cdot)$ denotes the convolution operation.

These streams are processed through successive stages of cross-scale fusion, involving both upsampling and downsampling operations to exchange information across branches. After each fusion stage, the refined features are processed by ARDB modules. This three-stage fusion process is repeated to progressively enhance the integration of multiscale features.

In ARDB, the residual-dense block comprises five cascaded convolutional layers. At each layer, the convolution operates on the current input together with the concatenation of all preceding layer outputs, thereby building multilevel information pathways and forming a dense feature-propagation chain. This design allows shallow cues to flow directly to deeper layers, effectively mitigating the detail loss commonly observed in conventional networks. Formally,

$$f_1 = \mathrm{Conv}\left(f_{\mathrm{in}}\right) \tag{7}$$

$$f_2 = \mathrm{Conv}\left(f_{\mathrm{in}} \oplus f_1\right) \tag{8}$$

$$f_3 = \mathrm{Conv}\left(f_{\mathrm{in}} \oplus f_1 \oplus f_2\right) \tag{9}$$

$$f_4 = \mathrm{Conv}\left(f_{\mathrm{in}} \oplus f_1 \oplus f_2 \oplus f_3\right) \tag{10}$$

$$f_5 = \mathrm{Conv}\left(f_{\mathrm{in}} \oplus f_1 \oplus f_2 \oplus f_3 \oplus f_4\right) \tag{11}$$

where $\mathrm{Conv}(\cdot)$ denotes the convolution operation, f_{in} is the input feature, and $f_i (i \in \{1, 2, 3, 4, 5\})$ are the intermediate (and final) outputs produced by the residual-dense connections.

At the end of each ARDB, we append the CA that automatically assesses the importance disparity across color channels. By dynamically reweighting the red, green, and blue channels, the network markedly improves the correction of underwater color casts. The CA is defined as

$$W = \sigma\left(\mathrm{Conv}\left(\mathrm{ReLU}\left(\mathrm{Conv}\left(\mathrm{Pool}_{\mathrm{avg}}\left(f_5\right)\right)\right)\right)\right) \tag{12}$$

$$f_{\mathrm{out}} = \left(W \otimes f_5\right) \oplus f_{\mathrm{in}} \tag{13}$$

where $\mathrm{Pool}_{\mathrm{avg}}(\cdot)$ denotes the average pooling, $\mathrm{Conv}(\cdot)$ denotes the convolution operation, $\mathrm{ReLU}(\cdot)$ is the rectified linear unit activation, $\sigma(\cdot)$ is the sigmoid activation, and W denotes the attention weight matrix.

By adopting a high-resolution preservation strategy together with a three-branch cross-scale interaction and fusion mechanism, the network achieves continuous bidirectional aggregation of fine details, textures, and global semantics across the parallel streams, effectively mitigating the detail attenuation commonly observed in multiscale fusion. After each fusion stage, ARDB that combines dense residual connections with channel attention is inserted to establish cross-layer feature

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propagation paths and to emphasize salient cues. This design provides an efficient and reliable solution for multiscale feature integration in UIE.

E. MSIF Module

The MSIF module plays a central role in CSG-Net by integrating multiresolution features extracted by the MFPCSI structure. Underwater images often contain targets with diverse spatial scales and semantic contexts, ranging from large-scale structural backgrounds to fine-grained object details. Directly merging such heterogeneous features without alignment may result in structural inconsistency and perceptual distortion. To address this, MSIF adaptively integrates features from multiple streams using both spatial and channelwise attention mechanisms.

As illustrated in Fig. 4, the MSIF module is composed of two key submodules: the attention feature fusion module (AFFM) and the ARDB. The AFFM module performs multistream attention fusion, while the ARDB enhances local feature representation and propagation.

1) Attention Feature Fusion Module: The goal of AFFM is to perform a weighted integration of multiscale features F_1, F_2, and F_3, corresponding to outputs from the MFPCSI's parallel branches at different resolutions. First, all input features are projected to a common dimensionality of 32 channels and then concatenated along the channel dimension

$$F_c = \text{cat}\left[F_1, F_2, F_3\right]. \tag{14}$$

A cascaded attention mechanism is then applied to capture spatial dependencies and interchannel relationships

$$W = \text{Conv}\left(\text{CS}\left(\text{CS}\left(\text{SELU}\left(\text{Conv}\left(F_c\right)\right)\right)\right)\right) \tag{15}$$

where Conv denotes the convolution operation, SELU is the self-normalizing activation function, and CS represents a sequence of convolution and SELU layers.

The attention map W is evenly split across channels to generate individual weights for the three input streams

$$W_1, W_2, W_3 = \text{Split}\left(W\right). \tag{16}$$

These weights are used to compute a fused representation via attention-weighted summation

$$F_f = W_1 \cdot F_1 + W_2 \cdot F_2 + W_3 \cdot F_3. \tag{17}$$

To refine the fused features, we apply a residual transformation via convolution blocks and add the result to the deepest stream

$$F_{\text{out}} = F_3 + \text{CSC}\left(F_f\right) \tag{18}$$

where CSC denotes a sequence of 3×3 convolution layers interleaved with SELU activations.

2) Attention-Refined Dense Block: To further enhance contrast, detail retention, and high-level feature transmission, we introduce ARDB as a refinement module after fusion. ARDB (see Fig. 4) combines dense connectivity and a lightweight channel attention mechanism to reweight informative regions.

The channel attention begins with global average pooling of the input feature F_{in}

$$f_{\text{avg}} = \text{AvgPool}\left(F_{\text{in}}\right). \tag{19}$$

This is followed by dimensionality reduction using 1×1 convolution, ReLU activation, and channel restoration

$$W = \sigma\left(\text{Conv}\left(\delta\left(\text{Conv}\left(f_{\text{avg}}\right)\right)\right)\right) \tag{20}$$

where σ is the sigmoid function and δ is the ReLU activation.

The final output is obtained by reweighting the input features

$$R = W \cdot F_{\text{in}} \tag{21}$$

where R represents the enhanced image.

In practice, four consecutive ARDB modules are used in the MSIF to progressively refine the fused representation. This design strengthens the network to suppress noise, preserve edge integrity, and enhance contrast in restored underwater images.

The MSIF module strengthens the network's ability to capture global–local interactions across scales and selectively enhances informative regions in degraded underwater images. AFFM learns attention-aware fusion weights to reconcile scale-specific semantics, while ARDB reinforces channelwise discriminative capability and reconstruction robustness. Together, they provide an effective mechanism for underwater feature integration and restoration. Overall, MSIF plays a vital role in integrating scale-diverse features and bridging the global–local enhancement objectives of CSG-Net.

F. Loss Function

To ensure both pixel-level accuracy and perceptual quality in UIE, we employ a composite loss function that jointly optimizes structural fidelity and detail preservation. Specifically, we adopt a combination of L_1 loss and the structural similarity index measure (SSIM) loss $\mathcal{L}_{\text{SSIM}}$ [41] for both stages of CSG-Net, encouraging the enhanced output to be both numerically accurate and visually consistent with the reference image.

The overall loss function is defined as a weighted sum of losses from the two stages

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SG}} + \lambda_1 \mathcal{L}_{\text{SE}} \tag{22}$$

where $\mathcal{L}_{\text{SE}}$ and $\mathcal{L}_{\text{SG}}$ denote the loss functions for the SE-Net and the SG-Net, respectively. λ_1 is a weighting factor empirically set to 0.5.

Each stage-specific loss function combines L_1 loss and SSIM loss

$$\mathcal{L}_{\text{SE}} = \mathcal{L}_1 + \lambda_2 \mathcal{L}_{\text{SSIM}} \tag{23}$$

$$\mathcal{L}_{\text{SG}} = \mathcal{L}_1 + \lambda_3 \mathcal{L}_{\text{SSIM}} \tag{24}$$

where $\lambda_2 = \lambda_3 = 0.5$ in all experiments.

1) L_1 Loss: The L_1 loss is used to minimize the absolute difference between the enhanced image and the ground truth (GT):

$$\mathcal{L}_1 = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \left|J\left(i, j\right) - R\left(i, j\right)\right| \tag{25}$$

where H and W denote the image height and width, and $J(i, j)$ and $R(i, j)$ are the pixel intensities at position (i, j) in the reference and enhanced images, respectively.

L_1 loss provides better robustness in noisy underwater conditions, where distortion and scattering often introduce

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high-frequency noise. Moreover, L_1 loss encourages sharp edge reconstruction and fine texture preservation, which are critical for recovering structural details in underwater scenes.

2) SSIM Loss: To further enforce perceptual consistency, we incorporate SSIM loss, which measures structural similarity between two images based on luminance, contrast, and spatial structure

$$\mathcal{L}_{\text{SSIM}} = 1 - \frac{(2\mu_J\mu_R + C_1)(2\sigma_{\text{JR}} + C_2)}{(\mu_J^2 + \mu_R^2 + C_1)(\sigma_J^2 + \sigma_R^2 + C_2)} \tag{26}$$

where μ_J and μ_R are the means; σ_J^2, σ_R^2 are the variances, and σ_{JR} is the covariance of the reference image J and the enhanced image R. C_1 and C_2 are small constants that stabilize the division.

SSIM loss emphasizes the preservation of spatial structures, which is essential in underwater images that often contain scene-critical objects such as coral, rocks, or man-made targets. As SSIM aligns well with human visual perception, it complements the pixelwise L_1 loss to produce more visually pleasing results.

By applying the loss function to the SE-Net and SG-Net stages, CSG-Net can decouple global structure learning from local detail refinement, ensuring that each component is trained to specialize in its respective task. The loss in the first stage $\mathcal{L}_{\text{SE}}$ helps the network capture coarse structural priors and global degradation patterns, while the loss in the second stage $\mathcal{L}_{\text{SG}}$ further fine-tunes the enhancement process for high-fidelity outputs.

IV. Experimental Results

We evaluate the performance of our proposed CSG-Net on four representative underwater image datasets: T90 and C60 of UIEB [25], USSC [26], and U45 [27]. To validate the effectiveness of our method, we conduct comprehensive comparisons with eight state-of-the-art UIE techniques, including both traditional prior- and deep learning-based methods. These include GDCP (TIP'18) [42], ULAP (RCM'18) [16], SMBL (TB'20) [43], LUWE (CVPR'20) [44], WaterNet (TIP'19) [25], UWCNN (PR'20) [23], URSCT (TGRS'22) [45], and Semi (CVPR'23) [46].

We present both qualitative and quantitative comparisons to demonstrate the advantages of CSG-Net in enhancing visual quality and structural consistency. In addition, we conduct a series of ablation experiments to evaluate the contribution of each module in our framework.

A. Implementation Details

Our network is trained on 800 image pairs randomly selected from the UIEB dataset [25]. We adopt the Adam optimizer with a batch size of 4 and a learning rate initialized at 0.0002. The learning rate follows an exponential decay schedule with a decay rate of 0.9 every 20 epochs. The model is trained for 200 epochs using PyTorch 1.12.0 on a single NVIDIA Tesla V100 GPU.

All input images are resized to 256×256 to ensure consistency across all training and evaluation procedures. The loss weights λ_1, λ_2, and λ_3 are set as described in Section III-F.

B. Comparison Methods

To ensure fair evaluation, we compare our model with eight state-of-the-art methods, categorized as follows. *Traditional Methods:* GDCP [42], ULAP [16], SMBL [43], and LUWE [44]. *Deep Learning-Based Methods:* WaterNet [25], UWCNN [23], URSCT [45], and Semi [46]. Among these, URSCT can only process fixed-resolution inputs of 256×256 pixels due to architectural constraints. For consistency, we resize all input images to 256×256 for both training and inference in comparison methods.

All deep learning baselines are retrained on the same UIEB training subset as our model. For traditional methods, we use either publicly available implementations or author-provided executable tools with default parameters. The evaluation metrics used include PSNR, SSIM, underwater image quality measure (UIQM), and UCIQE, which collectively assess color restoration, contrast enhancement, and structural fidelity.

C. Datasets

To comprehensively evaluate the performance and generalization ability of the proposed CSG-Net, we conduct experiments on four underwater image datasets: UIEB [25] (including subsets T90 and C60), UCCS [47], U45 [48], and SateHaze1k [49].

1) UIEB [25]: The UIEB dataset contains 950 real-world underwater images; 890 images have corresponding reference images and 60 challenging samples without GT. We randomly selected 800 paired images for training. The remaining 90 paired images are used for testing and are denoted as **T90**. The 60 unpaired challenging images are used for qualitative evaluation only and are denoted as **C60**

2) UCCS [47]: The underwater color correction subset (UCCS) comprises real underwater scenes captured near Zhangzi Island. It includes three subsets dominated by blue, blue–green, and green tones. Each subset contains 100 images, resulting in a total of 300 images across the UCCS dataset. These subsets are used to evaluate the robustness of enhancement methods across varying water color conditions.

3) U45 [48]: The U45 dataset is a widely used underwater enhancement benchmark that contains 45 unpaired images affected by various degradations, including low contrast, color distortion, and haze. U45 is mainly used to test visual quality under diverse real-world conditions.

4) SateHaze1k [49]: SateHaze1k contains 1200 paired samples of hazy remote sensing images and their corresponding clear GT. The dataset is stratified by haze severity into three levels (thin, moderate, and thick fog), enabling a rigorous evaluation of dehazing methods and the robustness of remote sensing imagery.

D. Evaluation Metrics

To quantitatively evaluate the performance of CSG-Net and the comparison methods, we use a comprehensive set of reference metrics, including full reference and nonreference metrics, in the four datasets described in Section IV-C.

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1) Full-Reference Metrics: For datasets containing paired GT (i.e., T90 in UIEB), we adopt the following full-reference metrics.

1) *Peak Signal-to-Noise Ratio (PSNR)* [50]: Measures the ratio between the maximum possible signal and the noise. Higher values indicate lower distortion and greater fidelity to the reference image.
2) *Mean Squared Error (mse)*: Captures the average squared difference between the enhanced and reference images. Lower values indicate better reconstruction accuracy.
3) *SSIM* [41]: Evaluates perceptual similarity in terms of structure, luminance, and contrast. Higher scores reflect better structural consistency.
4) *Multiscale SSIM (MS-SSIM)* [51]: Extends SSIM by computing structural similarity at multiple scales, offering a more holistic perceptual evaluation.

2) No-Reference Metrics: For all datasets (including those without GT, such as C60, UCCS, and U45), we also apply the following no-reference metrics.

1) *UIQM* [52]: Aggregates colorfulness (UICM), sharpness (UISM), and contrast (UIConM) to provide a unified, underwater-specific image quality score. Higher values indicate better perceived quality.
2) *IE* [53]: Measures the average amount of information present in the image. Higher entropy suggests richer details and color distribution.
3) *NIQE* [54]: A distortion-generic metric based on natural scene statistics. Lower natural image quality evaluator (NIQE) values correspond to higher perceived image quality.
4) *CEIQ* [55]: Evaluates enhancement performance based on five contrast-related features. Higher contrast-enhanced image quality (CEIQ) values indicate stronger and more natural contrast enhancement.

By jointly employing these full- and no-reference metrics, we ensure a comprehensive and multiperspective evaluation of image quality, covering distortion fidelity, structural consistency, and perceptual enhancement under varying underwater conditions.

E. Visual Comparisons

We conduct extensive qualitative comparisons across all datasets to evaluate the visual performance of the proposed CSG-Net in relation to state-of-the-art methods.

1) UIEB Dataset: As shown in Fig. 5, CSG-Net delivers superior visual quality across various types of underwater degradation. Traditional methods such as GDCP [42] and ULAP [16] struggle with color overcompensation, either introducing greenish or reddish tones. SMBL suffers from poor enhancement in blue-shifted regions due to red channel dependence. LUWE tends to overexpose or distort local structures. Among deep learning-based methods, WaterNet and UWCNN exhibit low contrast or unstable brightness. URSCT and Semi improve perceptual quality, but they may introduce color imbalances under extreme shifts. In contrast, CSG-Net effectively restores natural color balance and structure with minimal artifacts.

2) Highly Dynamic and Turbid Environments: Qualitative examples in Fig. 6 are drawn from the UCCS dataset, which captures real coastal scenes near Zhangzi Island under highly dynamic and turbid conditions. These scenes exhibit severe backscatter and spatially nonuniform attenuation caused by suspended particulates and rapidly varying illumination, resulting in pronounced blue/green color casts, reduced contrast, localized haze, and texture wash-out. UCCS further partitions scenes into blue, blue–green, and green subsets; the cases shown here emphasize challenging high-turbidity, time-varying settings that are representative of real deployments and constitute a stringent testbed for UIE methods.

Qualitative comparisons (see Fig. 6) show that our method produces the most faithful visual results among all competitors; our method effectively removes the blue/green color cast, restores a neutral white balance, and enhances both global and local contrasts without oversaturation while preserving edges and fine textures and suppressing residual haze. Consistently clear backgrounds and well-defined structures further attest to the perceptual advantages of our approach. Moreover, as summarized in Table I, our method achieves competitive performance across the entire benchmark suite-frequently attaining top-tier PSNR, SSIM, and UIQM scores-corroborating its robustness and generalization across diverse underwater conditions.

Fig. 7 shows the pixel distribution curves of the enhanced images versus the reference images in the highly dynamic and turbid environment. The red curve represents the result, and the blue curve represents the reference. The difference between the two curves reflects how closely the enhanced result aligns with the GT. A smaller gap indicates that the enhancement method can more effectively restore the degraded underwater image. From the visual results, our method [see Fig. 7(j)] produces the most natural colors and clear textures, avoiding oversaturation [as seen in LUWE (e) and Semi (i)] or underenhancement [as seen in GDCP (b) and ULAP (c)]. Quantitatively, the top-left corner of each image shows a quality metric (e.g., PSNR or UIQM). Our method achieves the highest score (**25.575**), surpassing all other approaches, indicating superior overall restoration.

Furthermore, the pixel intensity distributions demonstrate that the red curve of our method aligns most closely with the blue reference curve, suggesting that our method preserves the global luminance and color balance more accurately than competing methods. This indicates that our approach achieves better fidelity to the reference images, effectively enhancing both detail and color consistency in challenging underwater scenes. Our method achieves the best alignment with the reference distribution, indicating high-fidelity restoration.

3) Extreme Color-Bias Underwater Environments: Beyond generic highly dynamic or turbid settings, Fig. 8 further stratifies challenging color conditions into three extreme color-bias environments as follows.

1) *EGB*-Extreme Green-Biased High-Turbidity: where green dominance and severe backscatter induce strong, spatially nonuniform attenuation;

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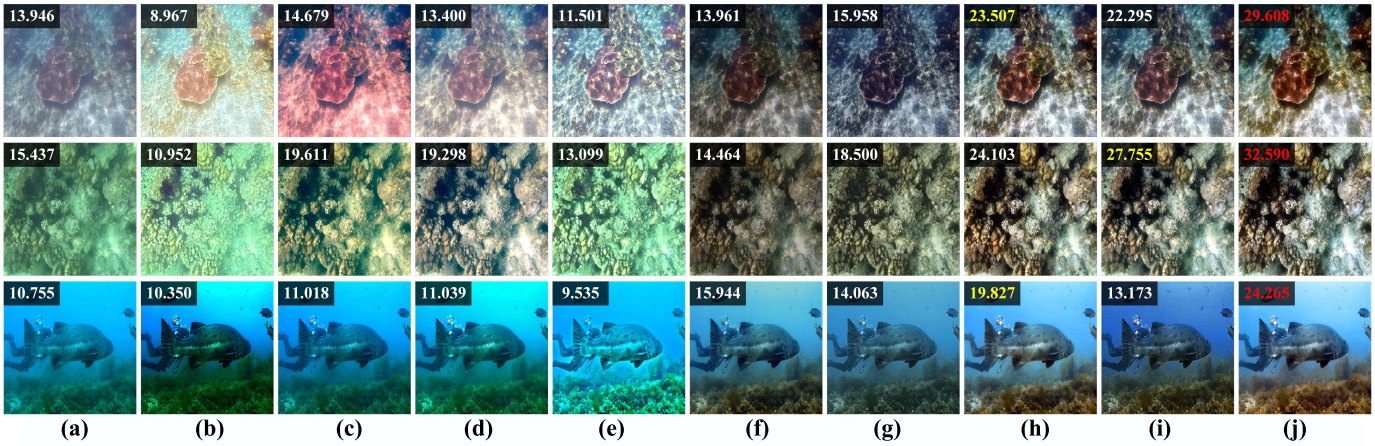

Fig. 5. Visual comparisons on representative samples from the T90 dataset, covering diverse underwater degradation types. PSNR values are shown in the top-left corner of each subimage. The red box indicates the best-performing method, while the yellow box denotes the second-best. (a) Raw. (b) GDCP. (c) ULAP. (d) SMBL. (e) LUWE. (f) WaterNet. (g) UWCNN. (h) URSCT. (i) Semi. (j) Ours.

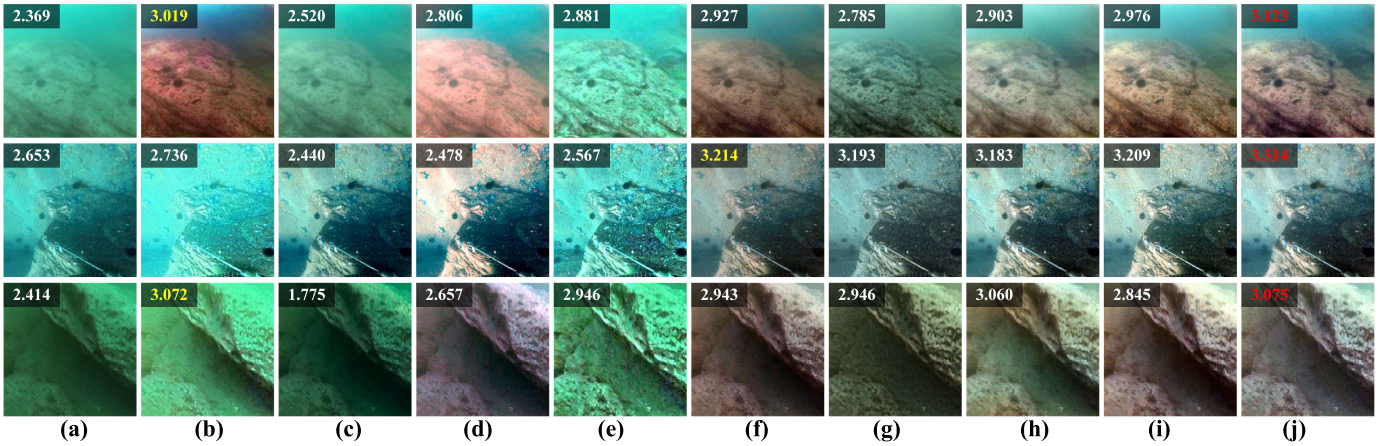

Fig. 6. Visual comparisons on representative samples with different degradation types from the UCCS dataset. The upper-left corner of each image shows the UIQM score. Red and yellow indicate the top-1 and top-2 performing methods, respectively. (a) Raw. (b) GDCP. (c) ULAP. (d) SMBL. (e) LUWE. (f) WaterNet. (g) UWCNN. (h) URSCT. (i) Semi. (j) Ours.

2) *EBB*-Extreme Blue-Biased Deep/High-Attenuation: characterized by heavy red-channel absorption and low global contrast; and
3) *BGMB*-Blue–Green Mixed-Bias Dynamic Turbidity: where blue/green dominance alternates across regions due to nonuniform illumination and localized turbidity. Across all three environments, our method achieves state-of-the-art visual fidelity in the reconstructions: our method consistently neutralizes blue/green casts, restores a balanced white point, and enhances both global and local contrasts without oversaturation while preserving edges and fine textures (see Fig. 8; quantitative trends are corroborated in Table I).

4) Multi-Illumination Composite Degradation Environments: To further assess the generalization of the proposed method, we also evaluate it under a heterogeneous multi-illumination composite degradation setting featuring mixed illumination, spatially nonuniform attenuation, and diverse surface reflectances; our method maintains strong color fidelity and structural detail in this environment as well (see Fig. 9).

This robustness stems from the complementary roles of the two stages: SE-Net encodes channel-dependent degradation priors and, together with the MSIF multistream fusion, stabilizes color rebalancing; MFPCSI preserves cross-scale structural fidelity, and ARDB in SG-Net refines fine-grained details. Moreover, leveraging an ImageNet-pretrained EfficientNet-B0 as an external prior-together with style-invariance constraints improves resilience to spatially nonuniform attenuation, strong backscatter, and localized turbidity, thereby strengthening cross-domain generalization.

5) Low-Light Enhancements: Fig. 10 illustrates enhancement results on low-light samples, along with corresponding brightness histograms. Most comparison methods either undercompensate (e.g., WaterNet [25] and Semi [46]) or overcorrect (e.g., ULAP [16] and LUWE [44]). CSG-Net produces results with balanced luminance and better visibility. The centered and symmetrical histogram further supports that our results avoid overbrightening or dark saturation.

6) Remote Enhancement in SateHaze1k Dataset: We evaluate the proposed method on remote sensing images from

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TABLE I

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE UIEB, UCCS, AND U45 DATASETS. THE BEST RESULTS ARE SHOWN IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Dataset	Metric	GDCP	ULAP	SMBL	LUWE	WaterNet	UWCNN	URSCT	Semi	Ours
UIEB-T90	PSNR ↑	13.386	15.913	16.597	13.812	17.349	17.985	22.092	23.554	**24.827**
	MSE ↓	0.228	0.174	0.159	0.212	0.144	0.134	0.085	0.073	**0.067**
	SSIM ↑	0.747	0.745	0.795	0.741	0.813	0.844	0.886	0.908	**0.919**
	MS-SSIM ↑	0.702	0.749	0.800	0.708	0.800	0.796	0.905	0.908	**0.918**
	UIQM ↑	2.670	2.259	2.543	2.846	2.917	3.011	**3.049**	2.989	3.012
	IE ↑	7.224	7.249	7.317	7.281	7.097	7.033	7.481	7.473	**7.526**
	NIQE ↓	6.413	5.751	5.856	6.856	5.580	5.887	5.596	**4.937**	5.404
	CEIQ ↑	3.227	3.237	3.314	3.275	3.107	3.106	3.414	3.414	**3.447**
UIEB-C60	UIQM ↑	2.110	1.639	1.904	2.598	2.399	2.514	2.609	2.566	**2.708**
	IE ↑	6.986	6.548	7.029	7.115	6.875	6.555	7.131	7.243	**7.315**
	NIQE ↓	6.358	5.868	6.231	6.319	5.921	6.055	5.808	5.688	**5.641**
	CEIQ ↑	3.131	2.820	3.147	3.189	2.991	2.853	3.223	3.303	**3.347**
UCCS	UIQM ↑	2.643	2.188	2.785	2.818	2.973	2.993	2.961	3.016	**3.022**
	IE ↑	6.954	6.701	7.165	6.995	7.198	7.083	7.158	7.303	**7.309**
	NIQE ↓	4.229	**4.011**	4.158	4.422	4.173	4.644	4.220	4.193	4.075
	CEIQ ↑	3.016	2.881	3.168	3.024	3.197	3.139	3.180	3.283	**3.284**
U45	UIQM ↑	2.275	2.287	2.388	2.482	2.957	3.064	**3.189**	3.122	3.125
	IE ↑	7.192	7.184	7.253	7.193	7.199	7.171	7.559	7.523	**7.571**
	NIQE ↓	4.120	4.275	4.132	**3.812**	4.549	4.080	3.910	4.038	3.874
	CEIQ ↑	3.191	3.191	3.249	3.198	3.186	3.213	3.453	3.438	**3.477**

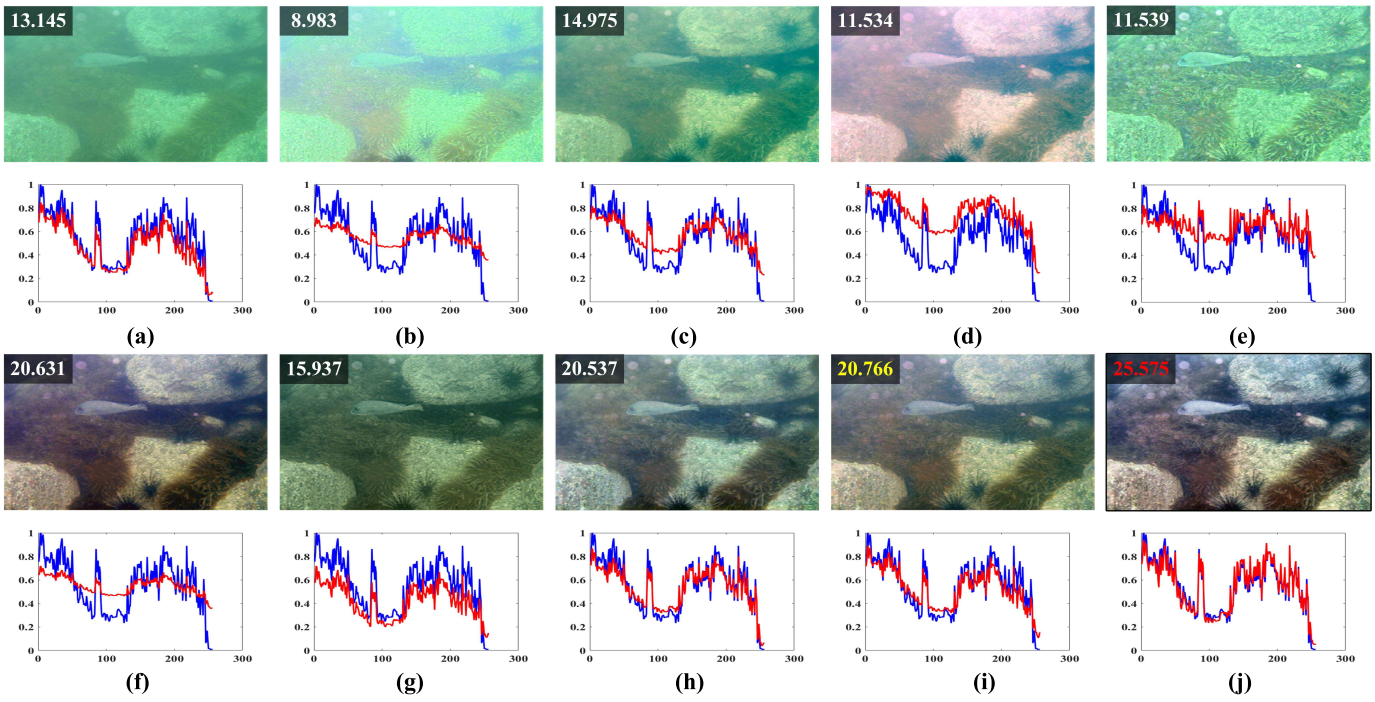

Fig. 7. Visual comparison and data distribution analysis on the highly dynamic and turbid environment. In addition to qualitative results, we analyze the pixel intensity distributions of enhanced images and their corresponding reference images. The red line denotes the distribution of the enhanced result, while the blue line represents the reference image. The proposed method exhibits the highest consistency with the reference distribution, indicating better restoration fidelity. (a) Raw. (b) GDCP. (c) ULAP. (d) SMBL. (e) LUWE. (f) WaterNet. (g) UWCNN. (h) URSCT. (i) Semi. (j) Ours.

the SateHaze1k dataset. Representative visual results are presented in Fig. 11. Traditional enhancement methods such as GDCP [42], ULAP [16], SMBL [43], and LUWE [44] fail to effectively handle certain challenging scenes. Traditional methods introduce additional haze artifacts in some cases, thereby degrading the visibility of remote sensing images. Deep learning-based methods demonstrate improved enhancement capabilities. However, residual haze is still evident in

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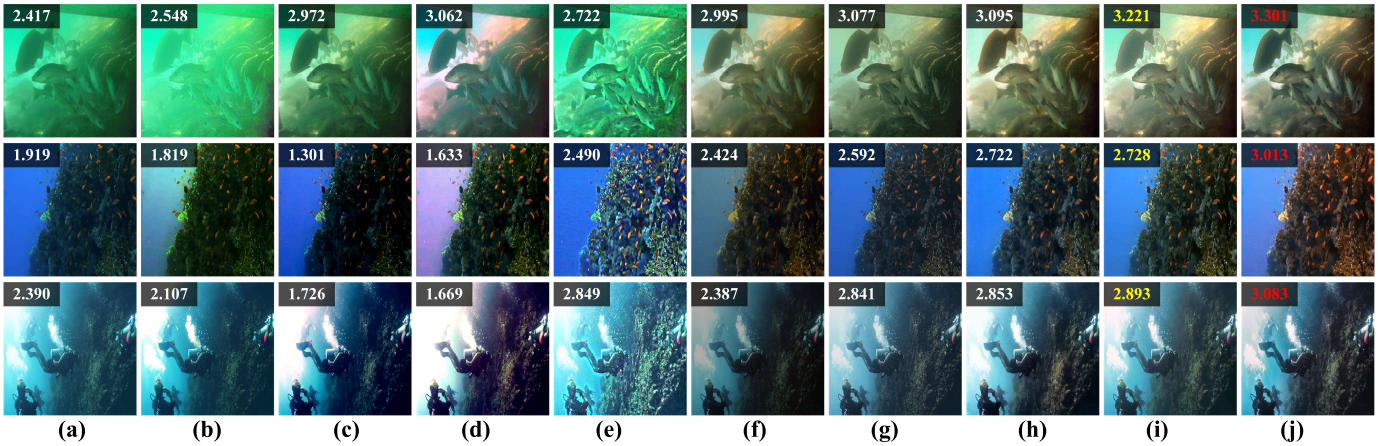

Fig. 8. Qualitative comparison on the C60 dataset under extreme color-bias environments. Row 1: extreme green-biased high turbidity with strong green casts and pronounced backscatter; row 2: extreme blue-biased deep/high attenuation characterized by heavy red-channel absorption and low global contrast; and row 3: blue–green mixed-bias dynamic turbidity where blue/green dominance alternates across regions due to spatially nonuniform illumination and localized turbidity. The upper-left corner of each image shows the UIQM score. Red and yellow indicate the top-1 and top-2 performing methods, respectively. (a) Raw. (b) GDCP. (c) ULAP. (d) SMBL. (e) LUWE. (f) WaterNet. (g) UWCNN. (h) URSCT. (i) Semi. (j) Ours.

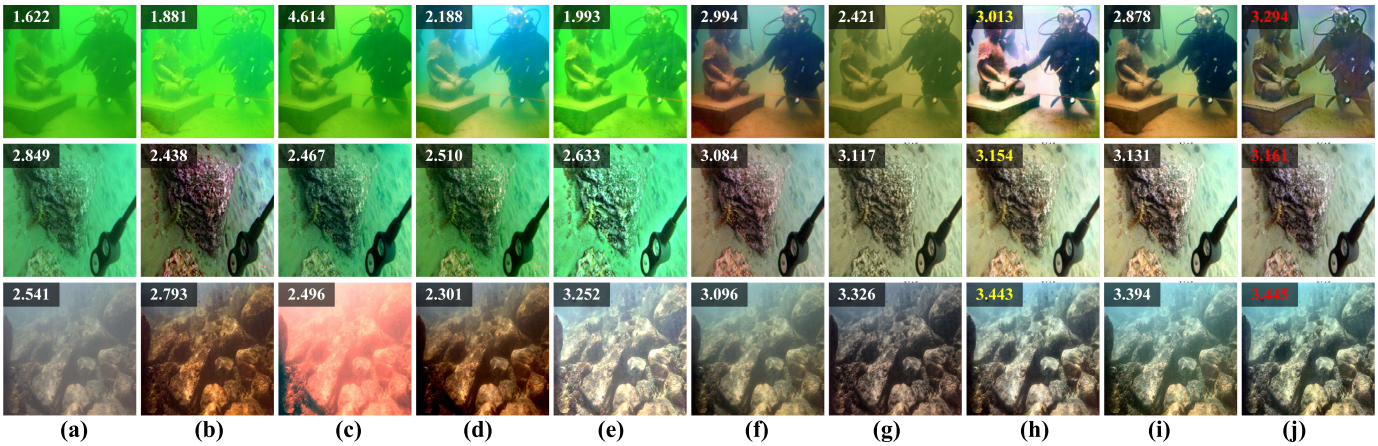

Fig. 9. Visual comparisons on multi-illumination composite degradation environments from the U45 dataset. The upper-left corner of each image shows the UIQM score. Red and yellow indicate the top-1 and top-2 performing methods, respectively. (a) Raw. (b) GDCP. (c) ULAP. (d) SMBL. (e) LUWE. (f) WaterNet. (g) UWCNN. (h) URSCT. (i) Semi. (j) Ours.

the enhancement results by UWCNN [23] and URSCT [45], especially in regions with complex textures. WaterNet [25] and Semi [46] perform better in haze removal. However, WaterNet [25] and Semi [46] exhibit limitations in recovering fine structural details, particularly in localized regions such as the rooftop area in the upper-right portion of the image.

In contrast, our method delivers markedly superior haze removal while simultaneously reconstructing sharper and more faithful structural details, ensuring robustness across both uniform regions and highly textured areas. The corresponding histogram analysis further supports this observation: the RGB histograms of our enhanced images exhibit the closest alignment with those of the GT, indicating superior consistency in both color distribution and image fidelity, which demonstrates the effectiveness of our method in producing visually and statistically superior enhancement outcomes.

7) Detail Recovery: In Fig. 12, we present zoomed-in comparisons to evaluate fine texture preservation. GDCP, LUWE, and UWCNN fail to recover structural details, while Semi and URSCT suffer residual color shifts. In contrast, CSG-Net demonstrates clearer textures, sharper edges, and accurate color restoration, contributing to natural and artifact-free enhancement.

F. Quantitative Comparisons

To ensure fair evaluation, all learning-based methods are retrained on the same training set, and all test images are resized to 256×256 to maintain consistency. We employ both full- and no-reference metrics, as detailed in Section IV-D.

Table I summarizes the quantitative performance across four benchmark datasets: T90, C60, UCCS, and U45. Our method consistently achieves top or second-best performance in nearly all metrics, demonstrating its robustness and adaptability across diverse underwater scenes.

On the UIEB-T90 dataset, CSG-Net significantly outperforms all competing methods in full-reference metrics (PSNR, SSIM, MS-SSIM, and MSE), with a clear margin. Although URSCT and Semi show slightly higher UIQM and lower NIQE

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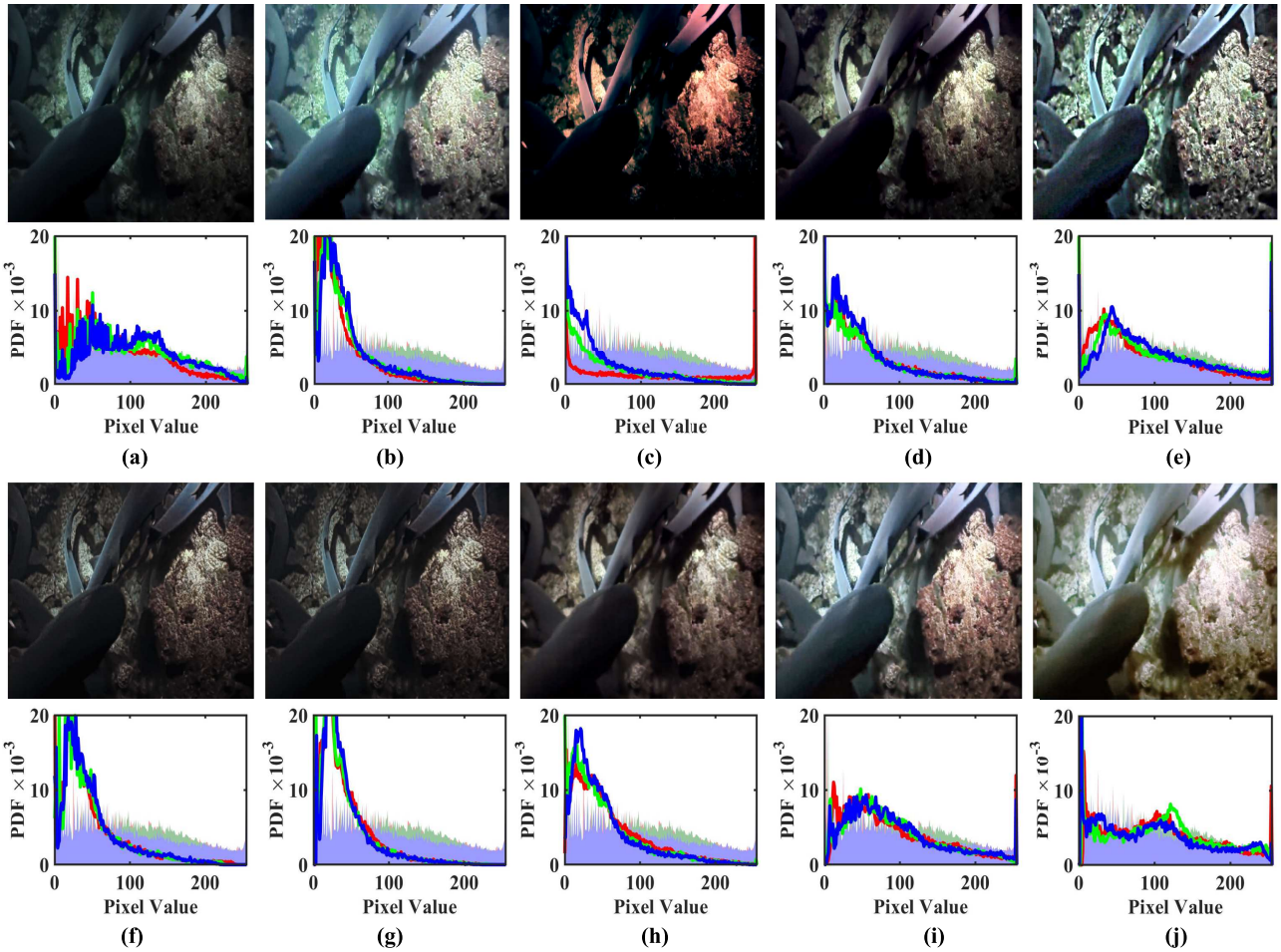

Fig. 10. Visual comparisons and corresponding brightness histograms on low-light samples from the T90 dataset. The bars represent the RGB histogram of the GT image, while the lines denote the corresponding RGB histogram distributions of the UIE image. (a) Raw. (b) GDCP. (c) ULAP. (d) SMBL. (e) LUWE. (f) WaterNet. (g) UWCNN. (h) URSCT. (i) Semi. (j) Ours.

scores, our results remain competitive and rank second in both metrics, indicating a strong balance between perceptual quality and structural fidelity.

For the challenging referenceless C60 subset, our method ranks first in all four no-reference metrics (UIQM, IE, NIQE, and CEIQ), showing superior generalization on low-quality and unpaired underwater images.

In the UCCS dataset, which includes samples with varied water tones, our method achieves the highest UIQM, information entropy (IE), and CEIQ scores. Although ULAP performs slightly better on NIQE, CSG-Net remains second best, demonstrating stable perceptual quality under real-world degradations.

For the U45 dataset, CSG-Net achieves the best IE and CEIQ values and ranks second in UIQM and NIQE. These results collectively validate our method's effectiveness in preserving contrast, enhancing structural details, and delivering perceptually pleasing results across multiple real-world conditions.

G. Complexity and Efficiency Analysis

To further demonstrate the practicality of the proposed CSG-Net, we conducted a comprehensive analysis of its complexity and efficiency compared with representative baseline methods. Specifically, we report the number of parameters (Params), floating-point operations (FLOPs), and inference speed in frames per second (FPS) under the same hardware environment. As shown in Table II, although our method does not have the smallest parameter count or GFLOPs, it attains the highest throughput at 25.27 FPS, notably faster than all competitors. We attribute the superior FPS to an efficiency-oriented design, EfficientNet-B0 backbone with depthwise-separable convolutions and squeeze-and-excitation for high compute density and good memory locality; MFPCSI+MSIF pipeline that performs cross-scale exchange at matched resolutions using 1×1 channel bottlenecks to keep fusion cost low; and ARDBs that employ lightweight attention within residual/dense paths, avoiding costly global attention or iterative refinement. This combination is highly parallelizable on GPUs, turning a moderate FLOP budget into high throughput. Overall, the results indicate a favorable accuracy–efficiency tradeoff and support the practicality of our model on resource-constrained platforms.

These results indicate that our method achieves a favorable tradeoff between efficiency and performance: the inclusion of the proposed modules significantly boosts visual quality (as evidenced by PSNR, SSIM, and UIQM improvements) while

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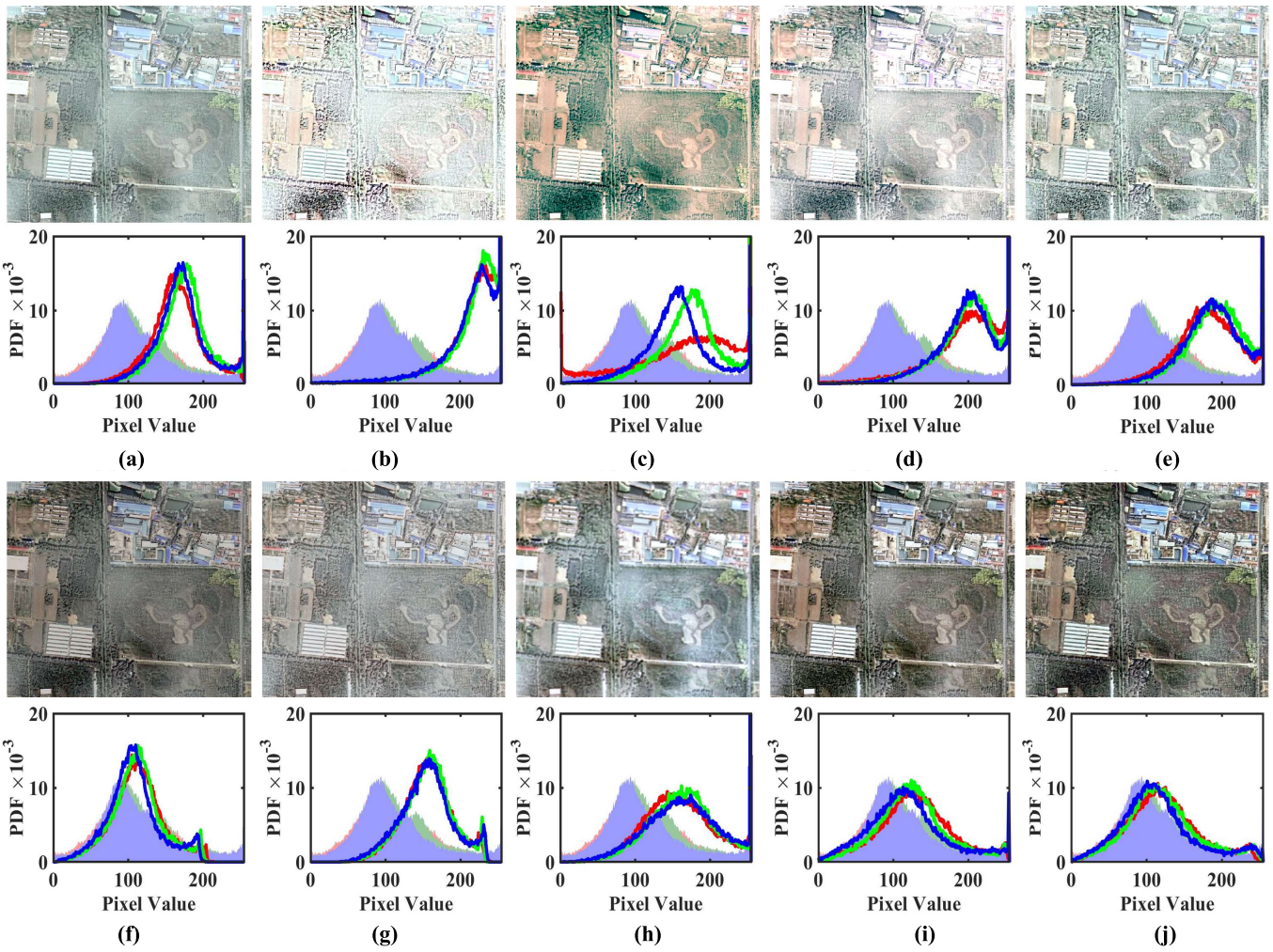

Fig. 11. Visual comparisons and corresponding brightness histograms on the remote image from the SateHaze1k dataset. The bars represent the RGB histogram of the GT image, while the lines denote the corresponding RGB histogram distributions of the UIE image. (a) Raw. (b) GDCP. (c) ULAP. (d) SMBL. (e) LUWE. (f) WaterNet. (g) UWCNN. (h) URSCT. (i) Semi. (j) Ours.

TABLE II

COMPARISON OF MODEL COMPLEXITY AND EFFICIENCY AMONG DIFFERENT METHODS. WE REPORT THE NUMBER OF PARAMETERS (PARAMS), FLOPS, AND INFERENCE SPEED MEASURED IN FPS UNDER THE SAME HARDWARE ENVIRONMENT. THE BEST RESULTS ARE SHOWN IN BOLD, AND SECOND-BEST RESULTS ARE UNDERLINED

Metric	GDCP	ULAP	SMBL	LUWE	WaterNet	UWCNN	URSCT	Semi	**Ours**
Params ↓	-	-	-	-	<u>1.291</u>	**0.04**	11.362	1.675	1.642
GFLOPs ↓	-	-	-	-	**1.170**	<u>5.230</u>	36.212	72.877	52.871
FPS ↑	6.13	2.79	0.33	0.22	1.01	4.11	<u>8.67</u>	3.24	**25.27**

introducing only limited computational overhead. Therefore, CSG-Net is not only effective in enhancing underwater images but also practical for deployment on resource-constrained platforms.

H. Quantitative Analysis Under Varying Training Ratios

To further evaluate the data efficiency of the proposed method, we trained our model using different proportions of the training dataset, including 25%, 50%, 75%, and 100% of the full data. As shown in Figs. 14 and 15, both PSNR and SSIM metrics consistently improve as the amount of training data increases, indicating that the proposed model effectively leverages the enhanced visual diversity and statistical richness provided by larger training sets.

Notably, even when trained with only 25% of the available data, the proposed model achieves 22.9-dB PSNR and 0.906 SSIM, which are already competitive with models trained on substantially larger datasets. This performance stability is mainly attributed to the pretrained EfficientNet backbone, which provides strong low-level feature representations and robust initialization for visual perception. By leveraging pretrained weights, the network can effectively generalize to unseen underwater conditions even under limited supervision.

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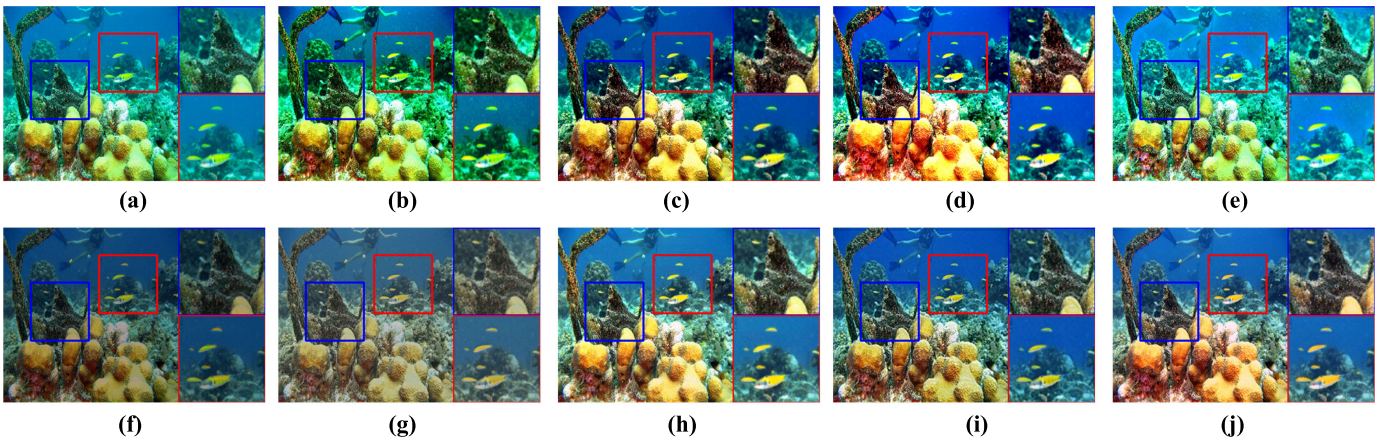

Fig. 12. Detail enhancement comparisons on representative samples from the T90 dataset. Zoomed-in regions highlight the ability of each method to preserve fine textures and correct color distortions. Our method produces sharper edges, richer textures, and more natural color rendering with minimal blurring and color cast. (a) Raw. (b) GDCP. (c) ULAP. (d) SMBL. (e) LUWE. (f) WaterNet. (g) UWCNN. (h) URSCT. (i) Semi. (j) Ours.

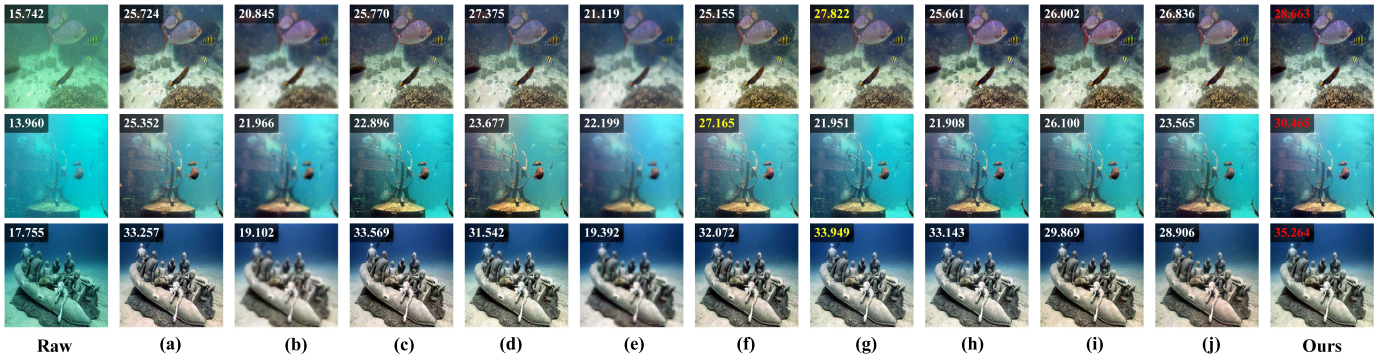

Fig. 13. Qualitative results of the ablation study on the UIEB-T90 dataset. The PSNR values in the upper-left corners indicate the restoration performance of each model variant. Red highlights the best-performing configuration, while blue denotes the second-best. The full CSG-Net framework produces images with the most natural color balance and enhanced structural details, validating the contributions of its components.

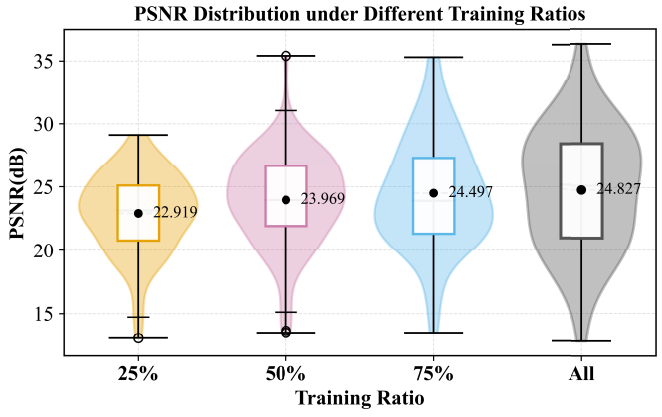

Fig. 14. PSNR distribution under different training ratios.

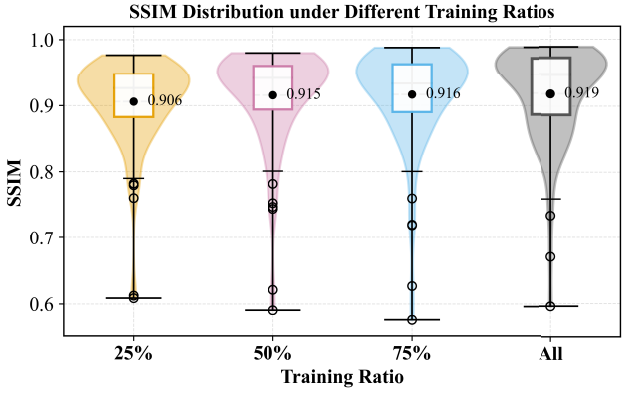

Fig. 15. SSIM distribution under different training ratios.

As the training ratio increases to 50%, 75%, and eventually 100%, both metrics continue to improve, reaching 24.8-dB PSNR and 0.919 SSIM at full data utilization. This demonstrates that while pretrained features provide a strong foundation, larger training datasets still play a crucial role in enhancing the network's domain adaptation and representation robustness.

Overall, these results validate that our method achieves high performance even under low-data regimes while maintaining scalability and consistent improvement as more training data become available.

I. Ablation Experiments

To assess the contributions of each component in CSG-Net, we perform ablation studies on the UIEB-T90 dataset.

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TABLE III
ABLATION STUDY RESULTS ON THE UIEB DATASET. × DENOTES THAT THE CORRESPONDING MODULE IS REMOVED. BOLD INDICATES THE BEST RESULT, AND UNDERLINED INDICATES THE SECOND-BEST RESULT

Module Removed		A	B	C	D	E	F	G	H	I	J	Ours
SE-Net	MSIF-ALL	×			×							
	AFFM		×	×				×	×			
	ARDB		×	×			×		×			
SG-Net	MSIF-ALL		×			×						
	AFFM	×		×				×		×		
	ARDB	×		×			×			×		
CA Block											×	
PSNR ↑		24.315	18.516	24.399	24.448	18.640	24.699	24.632	24.626	<u>24.726</u>	23.911	**24.827**
SSIM ↑		0.914	0.564	0.915	0.914	0.568	0.914	0.915	0.916	**0.921**	0.909	0.919

See Fig. 13 and Table III summarizes the results in terms of PSNR and SSIM.

1) Effect of the Dual-Stage Architecture: To evaluate the interdependence of SE-Net and SG-Net, we further introduce a variant that retains only SG-Net without global style priors. As shown in Table III (Model K), PSNR drops to 19.24 dB and SSIM to 0.582, indicating substantial degradation. In contrast, removing SG-Net while retaining SE-Net (Model E) results in a sharper performance drop, demonstrating that SE-Net facilitates global structure modeling, while SG-Net is essential for fine detail recovery. These findings highlight the strong complementary roles and synergistic benefits of the dual-stage design.

Model D (removing SE-Net) and Model E (removing SG-Net) achieve PSNR scores of 24.448 and 18.640, and SSIM values of 0.914 and 0.568, respectively. Compared to the full model (PSNR: **24.827** and SSIM: **0.919**), both models show performance drops, with Model E suffering a substantial decline of **25.0%** in PSNR and **38.2%** in SSIM. This demonstrates that both SE-Net and SG-Net are essential; SE-Net provides global structure restoration, while SG-Net refines local texture details.

2) Effect of the MSIF Module: Models A, B, and C remove MSIF in varying configurations. Model A (MSIF removed from both stages) shows a decrease of 24.315 (PSNR) and 0.914 (SSIM). Model B (removing only from SG-Net) drops significantly to 18.516 (PSNR) and 0.564 (SSIM), while Model C (removing all MSIF) records 24.399/0.915. These results indicate that the MSIF module in SG-Net plays a crucial role in structure and detail enhancement. Notably, removing MSIF from SE-Net alone (Model H) leads to a PSNR of 24.726, showing that SE-Net's MSIF contributes more modestly but still meaningfully to the final performance.

3) Effect of MSIF Submodules: AFFM and ARDB: Models F and G individually ablate AFFM and ARDB. Compared to our model (24.827/0.919), Model F achieves 24.699/0.914, and Model G produces 24.632/0.915. This indicates that both components contribute significantly: removing either leads to a **0.5**-dB PSNR drop and perceptible structural degradation in SSIM.

4) Effect of CA: Model J (removes the CA) results in a PSNR of 23.911 and an SSIM of 0.909. This corresponds to a **3.7%** drop in PSNR and **1.1%** in SSIM, confirming that CA helps improve feature discrimination by capturing channel dependencies.

5) Resolution Consistency Note: Since SE-Net outputs features at 1/4 resolution, we applied bilinear upsampling in Models B and E to ensure fair quantitative comparisons.

Table III summarizes the results of our detailed ablation study, conducted to better understand the role of each component. The most significant performance drop is observed in Model B, where the MSIF module is removed only from SG-Net. In this case, PSNR decreases drastically to 18.516 and SSIM to 0.564. This sharp decline highlights that the MSIF module within SG-Net is indispensable for cross-scale feature fusion and fine-grained detail recovery; its absence prevents the network from effectively integrating semantic and structural cues across resolutions, leading to blurred textures and amplified noise. A similarly large degradation occurs in Model E, where SG-Net is entirely removed, reducing the PSNR to 18.640. Although slightly less severe than Model B, this result further underscores that SG-Net plays a central role in enhancing local structures and restoring fine textures.

In summary, these observations reveal that SE-Net provides global degradation-aware constraints, whereas SG-Net is primarily responsible for preserving local details and fine structures. The complementarity of the two branches explains the strong performance of the full model, while the ablation results clearly demonstrate that SG-Net contributes the most to the PSNR improvement.

J. Application to Downstream Vision Tasks

To evaluate the practical utility of our enhancement method, we tested its performance on several representative downstream vision tasks, including keypoint detection, feature matching, and semantic segmentation.

1) Keypoint Detection: As shown in Fig. 16, we compare keypoint detection results on the original images and those enhanced by our method, as well as several baseline approaches. Our method consistently yields the highest number of detected keypoints, highlighted in red, indicating improved texture clarity and preservation of local features.

2) Feature Matching: Fig. 17 presents the feature point matching results. The original image pairs yield 60 and

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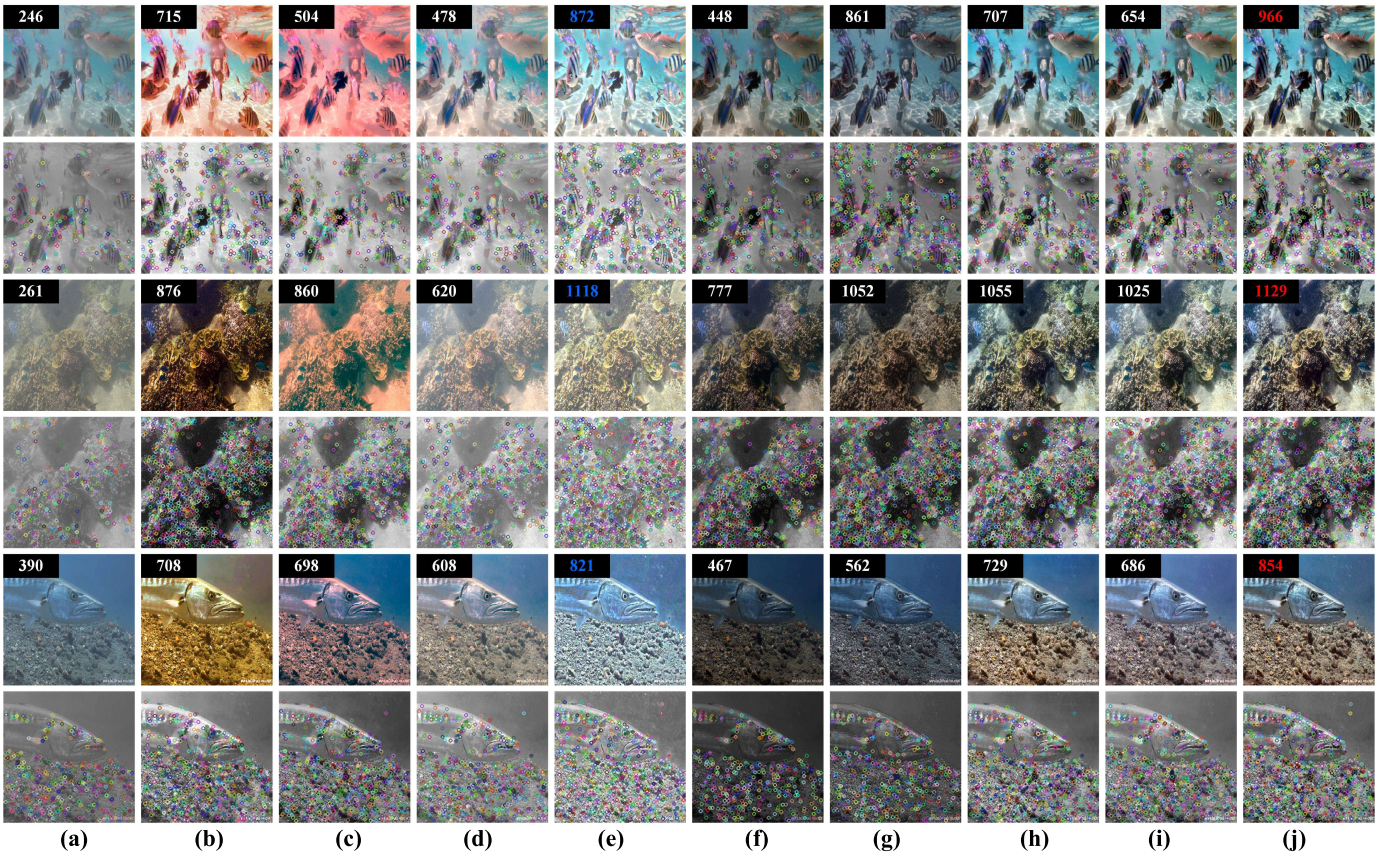

Fig. 16. Visual comparison of enhancement and keypoint detection results among different methods. The number of detected keypoints is indicated in the upper-left corner of each enhanced image. Red denotes the highest count, followed by blue. Our method preserves fine details and improves local contrast, facilitating more accurate and robust keypoint detection under complex underwater conditions. (a) Raw. (b) GDCP. (c) ULAP. (d) SMBL. (e) LUWE. (f) WaterNet. (g) UWCNN. (h) URSCT. (i) Semi. (j) Ours.

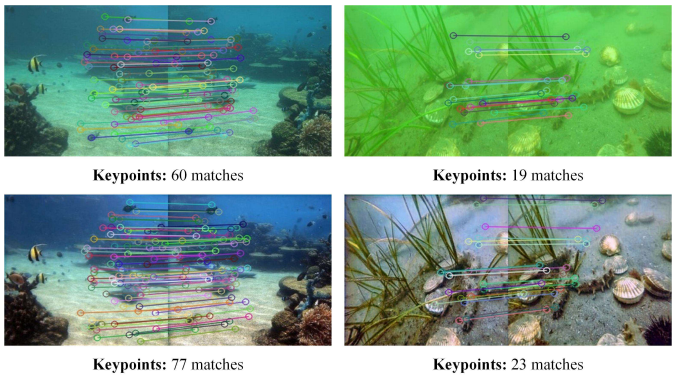

Fig. 17. Feature point matching results. The first row shows the original degraded underwater images, while the second row displays the results produced by our method. The improved visual quality leads to more accurate and dense feature correspondences, demonstrating the benefits of our enhancement model for downstream matching tasks.

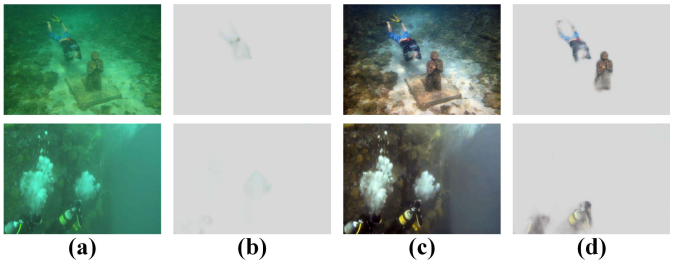

Fig. 18. Semantic segmentation results before and after enhancement. (a) Original underwater image. (b) Segmentation result on the original image. (c) Enhanced image by our CSG-Net. (d) Segmentation result on the enhanced image. Background regions are filled with gray for visual clarity. Compared to the original input, our enhanced results exhibit more accurate object boundaries and complete foreground segmentation under complex underwater conditions.

19 correct matches, while the enhanced counterparts using our method achieve 77 and 23 matches, respectively. These improvements demonstrate that our enhancement facilitates more reliable and dense correspondence estimation under challenging underwater conditions.

3) Semantic Segmentation: We further evaluate portrait segmentation performance before and after enhancement (see Fig. 18). The segmentation masks show that our enhanced images produce more accurate boundaries and complete region coverage, benefiting pixel-level understanding tasks in turbid underwater scenes.

These results confirm that our method improves perceptual quality and enhances the effectiveness of downstream high-level vision tasks. By restoring structural details and improving contrast, the proposed approach significantly supports tasks that require fine-grained features.

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V. Conclusion

In this study, we presented a CSG-Net enhancement network to tackle the challenges of UIE under complex physical degradations. Unlike prior approaches that either rely on handcrafted priors or monolithic learning models, our method explicitly decouples global degradation modeling and local detail enhancement via a two-stage architecture. The proposed style-guided mechanism enables the system to learn physically consistent degradation patterns and to adaptively restore color balance, contrast, and fine textures. Extensive experiments conducted on four widely used underwater image datasets demonstrate that CSG-Net achieves superior performance across both full- and no-reference metrics. Ablation studies further validate the effectiveness of each component, confirming that the style prior derived from SE-Net contributes significantly to global consistency, while SG-Net effectively enhances spatial detail. The MSIF module also proves essential in bridging multiscale semantic interactions. In addition, qualitative comparisons and improvements in downstream tasks (e.g., feature matching and segmentation) provide strong evidence that our design is theoretically grounded and practically impactful. Despite these promising results, challenges remain. The current framework focuses primarily on static degradation. Future work will explore the integration of temporal coherence and motion-aware modules to handle dynamic underwater scenarios, as well as the development of lightweight versions of CSG-Net for real-time deployment on resource-constrained underwater platforms.

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